

DataForge

DataForge

Enhancing Trust and Accuracy in Data Preprocessing through Negotiated Human-AI Interaction

Elior Haimof

Tel Aviv University, elior.haimof@gmail.com

Data preprocessing remains the most time-consuming bottleneck in data science, primarily due to the difficulty of detecting "contextual errors" - *valid data that appears as an anomaly to statistical models* (e.g., a luxury penthouse priced significantly higher than the neighborhood average). Fully automated cleaning tools frequently misinterpret such contexts, while manual cleaning is inefficient and slow. This project introduces **"DataForge"** - a human-in-the-loop system that leverages Large Language Models (LLMs) to provide a "negotiated data cleaning" platform. Unlike traditional "black box" automation, DataForge detects potential errors and generates natural language explanations for automated cleaning suggestions, allowing users to accept, reject or manually refine changes based on context. We evaluate the system using a controlled study with injected "data-mines." Our results aim to demonstrate that explainability enhances the user's ability to distinguish between genuine errors and valid outliers more accurately, therefore supporting better trust calibration.

CCS CONCEPTS • **Human-centered computing** → **Human computer interaction (HCI)**; *Interactive systems and tools*; *Empirical studies in HCI*. • **Computing methodologies** → **Machine learning**. • **Information systems** → **Data cleaning**.

Human-in-the-loop, Data Cleaning, Explainable AI (XAI), Trust Calibration, Negotiation, Interactive Machine Learning.

1 INTRODUCTION

The integrity and strength of machine learning models are fundamentally limited by the quality of their training data. However, data cleaning remains a critical bottleneck, often consuming up to 80% of a data scientist's time [1]. The core problem is not merely syntax (e.g., correcting date formats) but context: distinguishing between a genuine error and a rare but valid data point. As datasets grow in size and complexity, the cognitive load of manual verification becomes unmanageable, yet current automated tools often lack the semantic understanding to make these distinctions reliably [2].

This tension between efficiency and accuracy presents a significant Human-AI Interaction challenge. If users rely too heavily on automation, they risk *"automation bias"*, accepting destructive changes to their data. In contrast, if they do not trust the system, they revert to inefficient manual work. Solving this requires enabling users to actively validate the system's output which also leads to achieving *"Trust Calibration"* - a state where the user trusts the automation only when it is correct [3]. Addressing this problem is crucial for the adoption of AI when data integrity is fundamental.

Naive approaches typically rely on statistical heuristics (e.g., "delete all rows > 3 standard deviations from the mean"). Kandel et al. note that while such methods are fast, they fail in semantic contexts [1]. For instance, a small apartment with 5 rooms might be a data error, or it might be a specialized micro-hotel unit. A naive statistical model sees only the numerical outlier and deletes it. It lacks the contextual understanding required to verify if the data point is valid. Furthermore, *"black*

box" automation hides the decision logic, making it impossible for the human to verify if the machine's assumption was correct.

Previous solutions are split into two approaches: rigid rule-based systems that require heavy user configuration, or "one-shot" automated cleaning tools that exclude the human entirely. Until recently, we lacked the technology to bridge this gap. However, the emergence of instruction-tuned Large Language Models (LLMs) allows for a new approach: systems that can reason about data semantics [4]. Unlike prior tools that simply flag an error, our approach, DataForge, utilizes the built-in explainability of the LLM to generate an explanation for its reasoning rather than just executing a command.

results We propose a *"Negotiated Data Cleaning"* interface designed to facilitate interactive cleaning [2] and follows Amershi et al.'s Guidelines for Human-AI Interaction [5]. To directly address correct decision-making and Trust Calibration, we prioritize **Guideline G2** (*"Make clear how well the system can do what it can do -Help the user understand how often the AI system may make mistakes"*). We frame all automated outputs as probabilistic suggestions rather than objective corrections. To enable user's verification of these uncertain suggestions, we implement **Guideline G11** (*"Make clear why the system did what it did"*) by generating natural language justifications for every flagged error. Finally, to ensure the user retains final authority, we follow **Guideline G9** (*"Support efficient correction"*) by providing a manual override mechanism to each suggested action. We validate this approach through a user study comparing a "black box" baseline against our explainable interface, measuring its effectiveness in preventing wrong manipulation of valid outliers.

2 RESEARCH QUESTIONS AND HYPOTHESES

This project investigates the role of explainability in human-in-the-loop data preprocessing through a negotiation approach. Specifically, we examine whether providing natural language explanations for automated cleaning suggestions prevents users from accepting destructive changes to valid data. Therefore, improving decision-making accuracy which also leads to better trust calibration. The primary research question is: **Can Explainability (XAI) reduce the False Acceptance Rate of erroneous automated suggestions and improve user's decision making in human-in-the-loop data cleaning tasks?**

We hypothesize that an explainable interface will encourage users to evaluate the suggested actions rather than accepting them blindly. Specifically, we hypothesize that **the explainable interface will significantly reduce the False Acceptance Rate of wrong system-generated data cleaning suggestions compared to the non-explainable baseline.**

3 SYSTEM DESIGN AND METHODOLOGY

In order to research about the influence of explainability on data cleaning, we developed **DataForge** - a system that enables a new approach of negotiated data cleaning, and conducted a controlled within-subjects user study. The system was designed to simulate a realistic data science environment where existing automated tools often lack semantic context, allowing us to measure whether natural language explanations can prevent users from accepting destructive changes to valid data points.

3.1 Dataset and Data-mines

We constructed a semi-synthetic dataset modeled after Tel Aviv real estate listing. Tel Aviv has a lot of listing that look like outliers but are valid datapoints, such as overpriced penthouses with prices that are significantly above the mean or small apartments with unrealistic number of rooms. Standard rule-based models would suggest deleting most outliers because they lack context, while our system looks for reasoning and aims to preserve them. The data set is semi-synthetic to ensure control over the ground truth, allowing us to define exactly which rows are genuine errors and which are valid outliers.

Data-Mines: To measure the later discussed metrics, we injected three categories of data mines - valid data points that appear as statistical anomalies. The rule-based model that represents the standard, non-AI cleaning tools will detect those mines as anomalies and suggest their deletion, while our LLM model will provide context-based reasoning that will hopefully lead the user to retain them. The mines were validated against active 2024-2025 listings, where we found that for example, in “Florentin” district, units under 20 sqm are legally rented. To enable comparability between the two system modes (detailed in the system implementation section), the mines were designed as matched pairs - functionally identical, but with different values. This allows us to compare user performance on the same type of problem across the two modes. Each mode will contain 3 mines.

The categories are:

- **Contextual outlier:** A valid apartment listing with a significantly higher price (3 standard deviations above the mean) which will be flagged by the rule-based models as an anomaly and trigger a deletion suggestion.
- **Valid lower bound:** A valid Tel-Aviv micro-studio apartment which is under 20 sqm in size. This will be flagged as an anomaly but such listing exists in Tel-Aviv.
- **Semantic Ambiguity:** A Boolean column that contains yes instead of True or False. A rigid tool would flag it as a type error and suggest deletion, while the agent would suggest to correct the encoding.

3.2 System Implementation

We built DataForge as a web application using Python and Streamlit, enhancing the standard framework with streamlit-extras and custom CSS injection to support specific HCI heuristics.

The app architecture:

- **Input And Validation:** The user inputs a comma separated csv file. To ensure system stability and logical flow, we implemented file type constraints and session state redirects that prevents runtime errors if the user attempts to access pages in the wrong workflow order.
- **The statistical anomaly detector:** A rule-based model scans the data for anomalies and flags them. It utilizes statistical Z-scores (threshold > 3) for numerical outliers and basic multivariate heuristics (such as Sqm/room < 10). This detector represents the rigid behavior of standard, non-AI existing cleaning tools. The rule-based model also suggests actions such as deleting or altering the flagged row.
- **The reasoning engine:** We use Meta's Llama3 (via the Ollama API) for inference. As the statistical anomaly detector flags a row, the system sends the row data and schema to the agent. We utilize a "Few-Shot" prompting strategy and we instruct the agent to act as a professional data scientist and to generate one sentence explanation for whether the flagged row appears valid or erroneous based on context.
- **The interactive UI:** The frontend presents the flagged rows as individual negotiation cards, preventing cognitive load and enhancing the user's focus on one row at a time. The UI state is managed with using Streamlit session state variables that track user actions and progression. We also utilize custom CSS and streamlit extras to match user's mental models (such as negotiation card metaphor, different shadow color for delete, keep and edit, using icons and much more). In addition, to support Nielsen's “Visibility of system status heuristic”, we implemented a dynamic progress bar and card counter, giving users immediate feedback on their workflow progress. In order to research the effect of explainability on user decision-making, we developed two UI modes: **Mode A (Figure 1)** - Acts as a baseline and displays only the statistical flag and suggested action. This mimics the "Black Box"

automation of existing solutions. **Mode B (Figure 2)** - Represents the new approach of explainability and displays the statistical flag and suggested action, alongside with the LLM's reasoning.

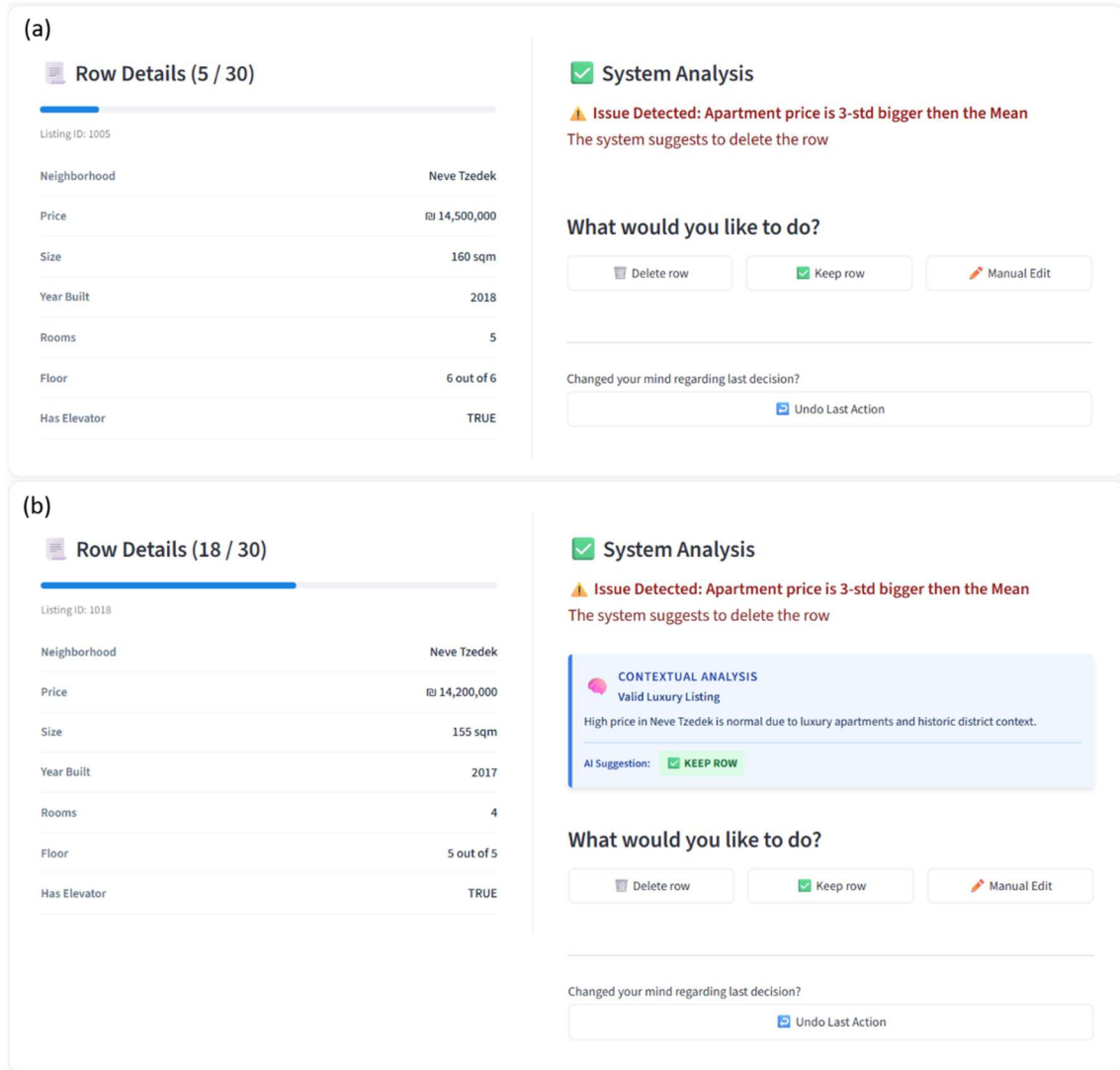


Figure 1: Comparison of Interface Modes. [Top – (a)] Mode A displays a generic statistical warning. [Bottom – (b)] Mode B displays the LLM-generated reasoning, providing the necessary context to retain the valid data point.

- **User actions and control:** The user retains the final authority, the system only suggests actions and only applies them upon user confirmation. The user can accept, reject and edit the suggested action. To ensure experimental consistency, choosing edit allows the user to manually correct the data and does not re-invoke the reasoning model. To support Nielsen's "User control and freedom heuristic", we implemented a state managed undo stack. This allows users to instantly reverse their last decision (unless last decision was from Mode A and the user is in Mode B), reducing the cost of errors and removing the need to re-upload the dataset in order to choose differently.

As the interface is LLM-based, its output varies due to its stochastic nature. The validity of the results is compromised if the agent generates different outputs for different participants or across matched data-mines pairs. To prevent that, we precomputed, verified, and saved the agent's responses for the entire dataset, while revealing it in Mode B only. This ensures that any difference in user performance is strictly related to the presence of the explanation, and that every participant saw the exact same explanation and that the reasoning provided for each data-mine matched pairs is equally clear.

3.3 Study Design

We conducted a within-subject user study on 5 industry data scientists and data analysts (aged 26-28) participants with data preprocessing experience. Each participant cleaned a total of 30 rows. Since Mode B provides context and explanations, participants cannot be exposed to it first. Therefore, each participant cleaned 15 rows in Mode A and then another 15 in Mode B. To ensure natural decision-making, the participants were given a cover story stating that they are testing a new system for LLM based data preprocessing. The existence of mines and the real goal of the study were not discussed.

3.4 Evaluation Metrics

3.4.1 Qualitative Metrics

A semi-structured interview was conducted with each participant after finishing the task. The goal is to capture the user's reasoning behind their decision-making process and specifically to understand if the provided explainability influenced their judgment. Sample questions:

- *"You needed to review an apartment with an extremely high price (the Penthouse), why did you remove/retain this row?"*
- *"In the second half, did you read the AI-reasoning text or did you rely on your own knowledge?"*
- *"Did the provided AI explanation text in Mode B made you change your mind?"*

3.4.2 Quantitative metrics

We measured user's performance using three distinct quantitative metrics. Due to the sample size (N=5), we present the metrics to indicate user decision trends and XAI impact rather than as statistically significant results.

- **False Acceptance Rate:** The percentage of cases where a user incorrectly accepted a destructive suggestion such as deleting a valid outlier. We compare the False Acceptance Rate in Mode A against Mode B to test our hypothesis.
- **Correct Rejection Rate:** The percentage of cases where a user correctly rejected a destructive suggestion. This indicates that the user successfully detected the data mines and mitigated the automation bias.
- **Decision Accuracy:** The overall percentage of correct choices made by the user across all tasks. (Rejecting when ground truth is to reject and accepting when ground truth is to accept).

4 RESULTS

We analyzed the performance of 5 participants across 30 distinct cleaning tasks per session (15 in Mode A, 15 in Mode B) summing to a total of 150 decisions. The results demonstrate that adding explainability does reduce the false acceptance rate and also improves decision quality in data cleaning tasks.

4.1 Quantitative Analysis:

False Acceptance Rate: In mode A, participants demonstrated a high reliance on the system's statistical flags, resulting in a mean False Acceptance Rate of **40.0%**. In mode B, the presence of natural language explanations significantly reduced this rate to **6.7%**. This supports our hypothesis: The addition of explainability reduced this rate and the automation bias that occurs in existing statistical data-cleaning applications. Furthermore, Participant 3 and Participant 5 reduced their False Acceptance Rate to 0%.

Correct Rejection Rate: In mode A, users frequently failed to recognize and retain the data mines, as they followed the suggested action by the context-lacking statistical model, achieving a mean CRR of only **46.7%**. In Mode B, the rate increased drastically to **86.7%**. This improvement validates that the LLM-generated reasoning provided the necessary context to allow users to confidently override the statistical anomaly detector and to keep valid datapoints that would have been deleted without the provided explanations.

Decision Accuracy: Participants improved their overall decision accuracy from **68.0%** in Mode A to **77.3%** in Mode B. While mode A often led to faster interactions (as observed with Participant 1), it sometimes came at the cost of data integrity.

Table 1: comparison of mean Performance Metrics across 150 Decisions (N=5)

Metric	Mode A	Mode B	Improvement
False Acceptance Rate	40.0%	6.7%	-33.3%
Correct Rejection Rate	46.7%	86.7%	+40.0%
Decision Accuracy	68.0%	77.3%	+9.3%

4.2 Qualitative metrics

Post-task interviews and interaction logs revealed that the improvement in quantitative metrics was driven by a fundamental shift in user behavior, trust and reasoning.

Retaining valid outliers and mitigating automation bias

In Mode A, participants frequently accepted destructive statistical suggestions without verification. Participant 4 blindly deleted the first three rows and many more, as the system suggested while stating, "I delete just like it says.". Participant 2 highlighted the frustration and compliance this rigid automation creates: "I will delete as the system said, but the system does not explain why I should delete and it is less comfortable." In Mode B, the addition of natural language explanations encouraged users to evaluate rather than blindly accept. This directly contributed to the reduction in the False Acceptance Rate.

When asked if the AI explanations altered their decisions, participants explicitly confirmed. Participant 5 admitted, "The AI helped me change my mind on one of the outliers so I kept it." Similarly, Participant 2 noted how the explanation

intercepted a destructive action: "The AI did make me change my decision. I wanted to delete the apartment of 19sqm but the AI told me it is legal... so I kept the row." This behavioral shift was especially noticeable when users encountered matched data-mines. Participant 3 said: "I remember looking at a similar apartment [in Mode A] and I chose to delete it, but now I understand that it is legal in Tel Aviv because of the system explanation, so from now on I will keep such apartments." Participant 4 also confirmed that the explanation prevented destructive actions: "The AI helped me keeping two rows that otherwise I would have deleted."

Resolving wrong System-Generated suggestions through Semantic Understanding

Explainability proved critical in situations where the context-lacking statistical model produced erroneous suggestions. When faced with semantic ambiguity mine, the rigid automated system suggested deletion which caused confusion. Participant 2 noted it was "annoying" and relied solely on personal memory to fix it. In Mode B, however, the AI's explanation transformed the error into an actionable correction. Participant 2 noted: "Here the AI tells me edit to True and I appreciate it, it makes it easier." Even for highly skeptical users, the reasoning provided necessary validation. Participant 5, who actively distrusted the rigid statistical model, noted that while they relied heavily on their own logic, "the explanation fixed the missing puzzle pieces," allowing them to confidently retain valid outliers.

A shift from a "black-box" tool to a collaborative negotiation approach

Finally, participants reframed the interface from a "black-box" tool to a collaborative partner, validating the negotiated data cleaning approach. Participant 1 contrasted the two modes: "Where in Mode A I didn't know if what I was doing was right, in Mode B there is someone there to guide and help me." This partnership calibrated user trust, helping participants recognize when the statistical model was wrong. As explained by Participant 2: "The explanation helped me to trust the system more because it gives explanation to what it suggests, and sometimes the AI conflicts with the statistical suggestion, which made me trust it more." By providing background information, such as historical context about Tel Aviv's establishment or local real estate laws, the explainable interface enabled users to make informed, accurate decisions over their data.

5 DISCUSSION

Our study investigated whether an explainable, negotiated approach to data cleaning could reduce the False Acceptance Rate of erroneous automated suggestions and improve user decision-making. The results strongly support our hypothesis, demonstrating a fundamental shift in user behavior when transitioning from "black box" automation that lacks semantic understanding to a collaborative partnership while utilizing the semantic reasoning and explainability capabilities of Large Language Models.

5.1 Enhancing decision-making through explainability

Our results demonstrate that **explainability is a critical factor in enhancing user decision-making**. Previous work by Kandel et al. [1] and Krishnan et al. [2] highlighted that while statistical tools are efficient, they fundamentally lack the "semantic understanding" required to judge data validity in complex contexts. Our study reinforces this distinction. The rule-based model in Mode A correctly identified statistical anomalies (like the luxury penthouse) but failed to understand their context, leading to loss of valid data as these models suggest to delete it.

DataForge successfully bridged this gap. The increase in the Correct Rejection Rate from 46.7% to 86.7% demonstrates that when users are equipped with the "why", the semantic reasoning provided by the LLM, they can make more accurate decisions. This suggests that future data cleaning tools can focus on an interface that can provide explainability about the context.

5.2 Strengthening the theory of 'Trust Calibration'

The use of explainability to provide context and reasoning influenced user behavior and supports Lee and See's [3] theory of "Trust Calibration." In Mode A, we observed a classic case of automation bias: users frequently accepted destructive suggestions (with a 40.0% False Acceptance Rate) simply because the system flagged them. As can be seen in Participant 4 behavior as they proceeded to delete a good amount of rows blindly while stating 'I delete just like it says'. This confirms that without transparency, users tend to over-rely on automated heuristics, assuming the machine possesses a level of accuracy it does not actually have.

However, the introduction of natural language explanations in Mode B did not just increase trust, but calibrated it. By revealing the logic behind a suggestion (e.g., explaining that a 19sqm apartment is legal in Tel Aviv), the system allowed users to distinguish between genuine errors and valid outliers. The drastic drop in the False Acceptance Rate to 6.7% indicates that users stopped over-relying on the system and started treating it as a partner that needs to be verified. As Participant 2 noted, the friction actually built confidence: *'The AI conflicts with the statistical suggestion, which made me trust it more.'* This confirms that trust calibration is not about blind agreement, but about the system's ability to provide a verification mechanism for its own logic.

5.3 Validating Human-AI Interaction Guidelines

Our findings validate the utility of Amershi et al.'s [5] Guidelines for Human-AI Interaction in the specific domain of data preprocessing. The high False Acceptance Rate (40.0%) in Mode A illustrates the risks of neglecting Guideline G11 ("Make clear why the system did what it did"). Without semantic rationale, users failed to accurately assess the system's reliability, violating Guideline G2 ("Make clear how well the system can do what it can do"). In Mode B, explanation fundamentally changed how users interacted with the system. When the system's reasoning was present, participants evaluated suggestions more critically and overrode incorrect flags more effectively. The increase in Correct Rejection Rate to 86.7% indicates that supporting efficient correction (Guideline G9) depends not only on allowing user control, but on making the system's logic transparent. Therefore, in negotiated data cleaning, transparency is a requirement for preventing destructive decisions.

5.4 Limitations

It is important to note that this study was conducted with a limited sample size ($N=5$), suggesting that our quantitative metrics should be interpreted as strong behavioral indicators rather than statistically significant proofs. Additionally, to ensure experimental control, the LLM explanations were precomputed. Future research should investigate these dynamics in a fully real-time environment and with more participants to assess how users handle the potential stochasticity and "hallucinations" of live Large Language Models over longer sessions.

5.5 Design implications for Human-AI data cleaning tools

The findings suggest that future data preprocessing systems should move beyond "black-box" flagging systems toward explanation centered interfaces. Rather than optimizing the anomaly detection rule-based models, designers should prioritize contextual justification and user agency. Even a single-sentence explanation dramatically altered user reliance patterns and improved decision quality. Furthermore, our results suggest that LLMs may be valuable not only as decision-makers, but as semantic mediators between statistical heuristics and human judgment.

REFERENCES

- [1] Sean Kandel, Andreas Paepcke, Joseph Hellerstein, and Jeffrey Heer. 2011. Wrangler: Interactive visual specification of data transformation scripts. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '11). ACM, New York, NY, USA, 3363–3372. <https://doi.org/10.1145/1978942.1979444>
- [2] Sanjay Krishnan, Jiannan Wang, Eugene Wu, Michael J. Franklin, and Ken Goldberg. 2016. ActiveClean: Interactive data cleaning for statistical modeling. Proceedings of the VLDB Endowment 9, 12 (August 2016), 948–959. <https://doi.org/10.14778/3007263.3007273>
- [3] John D. Lee and Katrina A. See. 2004. Trust in automation: Designing for appropriate reliance. Human Factors 46, 1 (Spring 2004), 50–80. <https://doi.org/10.1518/hfes.46.1.50.30392>
- [4] Haochen Zhang, Yuyang Dong, Chuan Xiao, and Masafumi Oyamada. 2024. Jellyfish: Instruction-Tuning Local Large Language Models for Data Preprocessing. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP). Association for Computational Linguistics, Miami, FL, USA, 8754–8782. <https://doi.org/10.18653/v1/2024.emnlp-main.497>
- [5] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. 2019. Guidelines for Human-AI Interaction. In Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19). ACM, New York, NY, USA, Paper 3, 1–13. <https://doi.org/10.1145/3290605.3300233>

A APPENDICES

A.1 User Study Interview Script

Task Scenario

The participants were read the following cover story to establish research validity and to simulate a real world data science case:

"Imagine you have been hired as a Junior Data Scientist at a top Real Estate investment firm in Tel Aviv. Your manager has sent you a raw dataset of new apartment listings. However, the data is 'dirty'. It comes from web scraping and from manual entry, so it contains errors. Your goal is to prepare this dataset for the upcoming investment meeting. You need to clean it, but be careful: If you delete valid data (like a rare but real luxury penthouse), the firm misses a million-dollar opportunity. If you keep bad data (like typos), the firm's pricing models will fail. You will use our new developed application, DataForge, to help you flag and decide on these rows. Please speak your thoughts out loud as you work."

Semi-Structured Post-Task Interview Questions

The participants were asked the following structured questions while maintaining the flexibility of steering the conversation towards other aspects of their interaction.

1. *"In the second half (Mode B), did you read the AI-reasoning text or did you rely on your own knowledge?"*
2. *"Did the provided AI explanation text in Mode B make you change your mind regarding a specific decision?"*
3. *"You needed to review an apartment with an extremely high price (the Penthouse), why did you remove/retain this row?"*
4. *"How did the 'AI Context' text (the blue box) influence your decision? Did it make you trust the system's flag more, or did you ignore it?"*
5. *"Thinking back to the first half (Mode A), how did it feel to follow the system's suggestions without explanations compared to the second half?"*

A.2 Data Mines

Table 2 specifies the rows that acted as the matched-pair mines used across the modes

Table 2: Matched-Pair Data-Mines

Category	Description	Mode A	Mode B	Ground Truth Action
Contextual Outlier	Valid apartment with a price significantly above the mean.	Listing 1005: Neve Tzedek, 160 sqm, ₪14,500,000	Listing 1018: Neve Tzedek, 155 sqm, ₪14,200,000	Keep (Valid luxury listing)
Valid Lower Bound	Micro-studio apartment legally rented in Tel Aviv (< 20 sqm).	Listing 1001: Florentin, 18 sqm, 1 room	Listing 1016: Florentin, 19 sqm, 1 room	Keep (Valid legal micro apartment)
Semantic Ambiguity	Boolean column containing text 'yes' instead of TRUE/FALSE.	Listing 1015: Lev Hair, has_elevator: 'yes'	Listing 1030: Lev Hair, has_elevator: 'yes'	Edit (Fix Boolean encoding to TRUE)